Debating Cosmological Models: Why ΛCDM Prevails Over Timescape Cosmology

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1. Introduction

Type Ia supernovae (SNe) and baryon acoustic oscillations (BAO) provide the most powerful probes of the universe's expansion history. While the standard ΛCDM model explains the observed acceleration of expansion through dark energy, alternative models such as the Timescape cosmology argue that acceleration may be apparent, arising from large-scale inhomogeneities and differences in gravitational time flow between cosmic voids and denser regions. This paper examines whether the Timescape model can account for key cosmological observables across epochs as effectively as ΛCDM. Models are evaluated on the basis of their ability to explain observed phenomena. The evidence from BAO–CMB–SNe coherence and consistency strongly favors ΛCDM as the superior model.

2. Observational Probes of Cosmic Acceleration: SNe and BAO

A comprehensive discussion of observational methods for detecting cosmic acceleration is given by Weinberg et al. (2013). Type Ia supernovae serve as "standard candles." Measuring their apparent magnitudes and spectral redshifts yields the luminosity distance–redshift relation,

$$d_L(z) = (1 + z)D_M(z), \tag{1}$$

where $D_M(z)$ is the comoving transverse distance. By comparing expected and observed luminosity distances, the relation requires acceleration $\ddot{a} > 0$.

BAO, in contrast, act as a "standard ruler." Galaxy surveys map the comoving scale r_d set by sound waves in the early universe. Angular and radial separations are related to observables via

$$s_\perp = D_M(z)\,\Delta\theta, \qquad s_\parallel = \frac{c\,\Delta z}{H(z)}. \tag{2}$$

Where $s_\perp$ is the transverse separation and $s_\parallel$ is radial. To convert observed quantities $(\Delta z, \Delta\theta)$ into physical distances, we must assume a fiducial cosmology. If our fiducial cosmology parameters are incorrect, $s_\perp$ and $s_\parallel$ will not match r_d. We then define ratios of correct scales to the fiducial sound horizon

$$\alpha_\perp = \frac{D_M(z)/r_d}{[D_M(z)/r_d]_{\rm fid}}, \qquad \alpha_\parallel = \frac{[H(z)\,r_d]_{\rm fid}}{H(z)\,r_d}. \tag{3}$$

These ratios are fit to measured BAO in DESI/BOSS to constrain Ω_m, H_0, and w (DESI Collaboration, 2025; Weinberg et al., 2013). Results from SNe and BAO are concordant—both indicating late-time acceleration. These results form the empirical basis for dark energy in the ΛCDM model.

3. Timescape's Argument

David L. Wiltshire and collaborators have argued that the apparent acceleration inferred from these observations could be an illusion arising from inhomogeneity. In this view, photons propagating from distant supernovae traverse voids that expand faster than the denser "wall" regions in which galaxies reside. Because clocks in these environments tick at different rates, an observer in a gravitationally bound region might perceive a differential redshift pattern that mimics cosmic acceleration, even though the global expansion is everywhere decelerating under gravity's tug. This hypothesis forms the basis of the *Timescape* cosmology.

4. The Timescape Cosmology and Apparent Acceleration

The Timescape model divides the universe into two idealized domains: expanding, negatively curved *voids* and spatially flat, slowly expanding *walls*. The fraction of total volume in voids is denoted by $f_v(t)$, which evolves according to (Wiltshire, 2007)

$$\frac{\dot{f}_v}{3(1-f_v)} = f_v(H_v - H_w), \tag{4}$$

where H_v and H_w are the local expansion rates of voids and walls respectively.

4.1. The Lapse Function and Clock-Rate Differences

Differences in regional expansion generate a lapse function that relates the global average ("bare") time to the proper time measured by observers in wall regions:

$$dt_{\text{wall}} = \bar{\gamma}^{-1}(t)\, dt_{\text{bare}}, \qquad \bar{\gamma}(t) = \frac{1 - f_v/2}{1 - f_v}. \tag{5}$$

As voids come to dominate the cosmic volume ($f_{v0} \approx 0.76$), this lapse grows significantly, producing a present-day wall–void age difference of approximately 3.3 Gyr. Wall observers thus measure a younger universe ($\tau_{w0} \approx 14.2$ Gyr) than the global average ($t_0 \approx 17.5$ Gyr).

4.2. Apparent Acceleration from the Model's Geometry

The volume-average scale factor $\bar{a}(t)$ always decelerates ($\ddot{\bar{a}} < 0$) under gravity's weight, but the effective "dressed" scale factor inferred by wall observers is

$$a_{\text{wall}}(t_{\text{wall}}) = \bar{\gamma}^{-1}(t)\, \bar{a}(t), \tag{6}$$

and its second derivative with respect to wall time,

$$\frac{d^2 a_{\text{wall}}}{dt_{\text{wall}}^2} = \bar{\gamma}^2 \left(\ddot{\bar{a}} - \frac{\dot{\bar{\gamma}}}{\bar{\gamma}} \dot{\bar{a}} \right), \tag{7}$$

can be positive even when $\ddot{\bar{a}} < 0$. This yields an *apparent acceleration* identical in form to what is inferred from SNe data—without introducing a cosmological constant. However, the magnitude of the 3.3 Gyr lapse emerges directly from the model's internal geometry, not from observational data on the actual gravitational potential or void network.

The lapse function in the Timescape cosmology is fully determined by the void fraction. Because the regional expansion rates satisfy $H_v > H_w$, the relative clock rate between wall observers and the volume–average time is given by

$$dt_w = \bar{\gamma}^{-1}(t)\, dt_{\text{bare}}, \qquad \bar{\gamma}(t) = \frac{1 - f_v(t)/2}{1 - f_v(t)}. \tag{8}$$

Thus, the magnitude of the clock–rate differential depends directly on the void volume fraction $f_v(t)$. The amount of apparent acceleration inferred from Type Ia supernova luminosity distances comes from this lapse function: only for a present–day void fraction $f_{v0} \approx 0.76$ does the model produce sufficient differential expansion for wall observers to infer $\ddot{a}_{\text{wall}} > 0$. The corresponding ~ 3.3 Gyr difference between the bare age of the universe and the proper time measured in wall regions is therefore not an independent prediction of the Timescape scenario, but a direct consequence of choosing the void fraction required to fit the supernova data. This connection is internal, in contrast with the ΛCDM framework in which supernovae, BAO, and CMB constraints arise from distinct physical effects yet independently find on the same cosmological parameters.

5. Observational Fits of the Timescape Model

Timescape cosmology has been compared against supernova observations in several analyses. Early fits by Wiltshire (2008) and Smale and Wiltshire (2011) showed that for $f_{v0} \approx 0.76$, the model reproduces the observed luminosity–distance relation almost as well as ΛCDM. More recently, Seifert et al. (2024) reported strong Bayesian evidence ($\ln B > 5$) favoring Timescape over ΛCDM when applied to the Pantheon+ sample with model-independent light-curve standardization. However, this analysis considered supernovae alone and did not include a self-consistent BAO or CMB calibration, leaving the broader cosmological fit undetermined.

6. CMB–BAO Coherence and the Success of ΛCDM

The coherence between CMB and BAO distance scales is one of the strongest arguments in favor of the ΛCDM model. The CMB measures the angular acoustic scale,

$$\theta_* = \frac{r_s(z_*)}{D_M(z_*)},$$

which represents the angular size of the sound horizon at photon decoupling (Weinberg et al., 2013; Planck Collaboration, 2020). Galaxy surveys measure the same physical scale at lower redshift through the transverse and radial BAO quantities

$$\alpha_\perp = \frac{D_M(z)/r_d}{[D_M(z)/r_d]_{\rm fid}}, \qquad \alpha_\parallel = \frac{[H(z)\, r_d]_{\rm fid}}{H(z)\, r_d},$$

which together trace the late–time expansion history of the Universe (Weinberg et al., 2013; DESI Collaboration, 2025). DESI DR2 and other recent surveys find remarkable agreement between these observables across cosmic time, reinforcing a single, self-consistent expansion history. When the Timescape model is tuned to reproduce the CMB peaks (Nazer and Wiltshire, 2015), it predicts $H_0^{\rm TS} \simeq 61$ km s^{-1} Mpc^{-1} and $f_{v0} \simeq 0.63$, which fail to match the observed BAO distances. No full Timescape fit to BAO data has yet been achieved, whereas ΛCDM naturally reproduces both CMB and BAO relations with the same parameters.

6.1. Calibrating the Sound Horizon and Predicting Late–Time Geometry

The success of ΛCDM arises from the way its parameters, determined from the CMB, are self-consistent with measurements in later phenomena. *Planck*'s measurements of the CMB power spectrum constrain the baryon and dark-matter densities, the expansion rate, and the optical depth, yielding a precise angular acoustic scale $\theta_* = r_s(z_*)/D_M(z_*)$. This calibration fixes the comoving sound horizon r_s and, by extension, the sound horizon at the drag epoch $r_d \approx r_s(z_d)$. Because the same baryon–photon physics sets both scales, r_d becomes a derived—not free—quantity within the model.

With r_d established, ΛCDM predicts how cosmic distances evolve with the dynamics of $a(t)$. The resulting functions $D_M(z)$ and $H(z)$ can then be compared directly with BAO and SNe analyses.

6.2. Matching BAO and Supernova Data

Large galaxy surveys measure the BAO feature in both transverse and radial directions, providing the quantities

$$\frac{D_M(z)}{r_d}, \qquad H(z)\, r_d,$$

which describe, respectively, the angular diameter distance and the expansion rate at different redshifts. Within the ΛCDM framework, these observables are not independently fitted at each redshift but are *predicted* from the same set of parameters—such as Ω_m, Ω_Λ, and H_0—that are first constrained by the CMB power spectrum. Once the model reproduces the acoustic peak structure of the CMB and therefore the correct ratio $\theta_* = r_s(z_*)/D_M(z_*)$, it automatically fixes the sound horizon r_d and the background evolution of the scale factor $a(t)$. This allows precise predictions for the redshift dependence of both $D_M(z)$ and $H(z)$, which can then be directly compared to BAO distance measurements from surveys like DESI and BOSS.

Type Ia supernovae provide a complementary test of the same expansion history. Their luminosity distances depend on integrals of $H(z)$ over redshift, meaning that the supernova Hubble diagram traces the same background evolution predicted by the ΛCDM parameters calibrated by the CMB. The excellent agreement between the observed SNe Ia distance–redshift relation and the ΛCDM predictions derived from CMB and BAO constraints demonstrates the model's internal consistency across independent cosmic phenomenon (Weinberg et al., 2013).

Although these predictions rest on the assumptions of the ΛCDM framework—spatial flatness, cold dark matter, and a cosmological constant—those assumptions allow the model to be fully constrained by early-universe data. Once the parameters are fixed, the same theoretical structure yields remarkably accurate predictions for independent observables, including late-time BAO distances and supernova luminosities. The fact that a single, self-consistent parameter set reproduces the CMB acoustic peaks, BAO measurements, and the SNe Ia luminosity-redshift relation is an impressive show of the ΛCDM model's predictive power.

7. Backreaction, Local Homogeneity, and Observational Constraints

To dive further into the dynamics of the Timescape model, it is useful to introduce the Buchert-averaged analog of the Raychaudhuri and Friedmann equations. For non-rotating dust, Buchert's averaging of Einstein's field equations over a spatial domain D yields

$$3\frac{\ddot{a}_D}{a_D} = -4\pi G\langle\rho\rangle_D + \Lambda + \mathcal{Q}_D, \tag{9}$$

where $a_D(t)$ is the volume scale factor of the domain and $\mathcal{Q}_D$ is the backreaction term,

$$\mathcal{Q}_D = \frac{2}{3}\big(\langle\theta^2\rangle_D - \langle\theta\rangle_D^2\big) - 2\langle\sigma^2\rangle_D. \tag{10}$$

Here, σ^2 is the shear scalar. For a pressureless cosmological fluid (dust), the quantity describing its local expansion is the scalar

$$\theta \equiv \nabla_\mu u^\mu, \tag{11}$$

the covariant *divergence* of the fluid four–velocity u^μ. This scalar measures the rate at which the volume of a comoving fluid element changes along the flow. The evolution of θ for any set of neighboring worldlines is governed by the Raychaudhuri equation,

$$\dot{\theta} = -\frac{1}{3}\theta^2 - 2\sigma_{\mu\nu}\sigma^{\mu\nu} + 2\omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}u^\mu u^\nu, \tag{12}$$

where $\sigma_{\mu\nu}$ and $\omega_{\mu\nu}$ are the shear and vorticity tensors. For an irrotational dust fluid ($\omega_{\mu\nu} = 0$ and $p = 0$), the four–acceleration vanishes and the matter content satisfies $T_{\mu\nu} = \rho\, u_\mu u_\nu$. Inserting this into the Einstein field equation gives

$$R_{\mu\nu}u^\mu u^\nu = 4\pi G\,\rho\,, \tag{13}$$

The Raychaudhuri equation then becomes

$$\dot{\theta} = -\frac{1}{3}\theta^2 - 2\sigma^2 - 4\pi G\rho, \tag{14}$$

showing explicitly how matter density and shear contribute to the local deceleration of the fluid. In an exactly homogeneous FLRW universe, the shear vanishes and $\theta = 3H$, reproducing the familiar Friedmann evolution, while in inhomogeneous cosmologies spatial variations in θ quantify departures from uniform expansion. For $\mathcal{Q}_D$ to drive an apparent acceleration in $a_D(t)$ without dark energy ($\Lambda = 0$), the variance of the expansion rate $Var_D(\theta) = (\langle\theta^2\rangle_D - \langle\theta\rangle_D^2)$ must be extremely large, which requires that different regions of the universe must expand at substantially different rates. This is best shown by relating the expansion scalar to the local Hubble rate by $\theta(\mathbf{x}) = 3H(\mathbf{x})$, then the variance term in the backreaction $\mathcal{Q}_D$ is directly tied to spatial fluctuations in H. Writing $H(\mathbf{x}) = \bar{H}\,[1 + \delta_H(\mathbf{x})]$ with $\bar{H} \equiv \langle H\rangle_D$, it follows that

$$\mathrm{Var}_D(\theta) = \langle\theta^2\rangle_D - \langle\theta\rangle_D^2 = 9\,\bar{H}^2\,\langle\delta_H^2\rangle_D, \tag{15}$$

so that the variance of the expansion scalar is directly proportional to the rms fractional deviation of the local expansion rate from the average over the domain. Thus, a large contribution to $\mathcal{Q}_D$ requires $\frac{\delta H}{H} \approx 0.3 - 0.5$ fluctuations in $H(\mathbf{x})$. This requirement of large inhomogeneities underlies the Timescape model, in which rapidly expanding, negatively curved voids occupy most of the cosmic volume, while wall regions expand more slowly, producing a significant lapse between void and wall clocks.

Ishibashi and Wald (2006) argue that such large backreaction effects are incompatible with the observed near-FLRW nature of our universe. The strongest evidence comes from the cosmic microwave background, where temperature anisotropies are at the level of $\delta T/T \sim 10^{-5}$. These tiny fluctuations imply extremely small perturbations in the gravitational potential and therefore very small spatial variations in the expansion scalar θ. Because the CMB probes the universe on the largest observable scales, any cosmological model requiring large differences in regional expansion rates is automatically in tension with observed isotropy.

Low-redshift observations reinforce this conclusion. BAO measurements make separate determinations of the radial and transverse distance scales. Although BAO analyses use a fiducial cosmology to convert angles and redshifts into distances, the *anisotropy* of the BAO feature—the ratio of radial to transverse clustering scales—is an observable that does not depend on the fiducial choice. If the true universe expanded at substantially different rates in different regions, as required by the Timescape scenario, the BAO peak in the twodimensional correlation function would appear physically distorted: the radial and transverse BAO scales would differ at the tens-of-percent level, producing an elliptical rather than circular BAO structure. DESI and BOSS instead find that the BAO feature is isotropic at the percent level, directly constraining the variance in the expansion rate and ruling out the large void–wall expansion differences needed for a dynamical $\mathcal{Q}_D$.

Similarly, the Hubble expansion rate is remarkably isotropic once peculiar motions are subtracted. Differences in Hubble rates between void and wall regions inferred by Timescape's model would be distinct and measurable when looking at void-dominated or wall-dominated regions of the sky. However, surveys measure Hubble expansion rate anisotropies $\frac{\delta H}{H}$ at less than 1% on cosmological scales, far below the 30–50% variation in expansion rate assumed between void and wall regions in Timescape. Together, these observations indicate that the actual variance in the expansion scalar, $\mathrm{Var_D}(\theta)$, is far too small for $\mathcal{Q}_D$ to become dynamically significant.

8. Discussion: Evaluating Competing Frameworks

Although Timescape successfully fits certain datasets and it does not seem presently possible to falsify the idea that apparent acceleration may arise from void–wall clock differences, ΛCDM achieves a far broader success in predicting observed phenomena—the basis for evaluation of the two models. CMB and BAO constraints compel Timescape to adopt "dressed" parameters that are inconsistent with measured Hubble rates and distances. Although Timescape theorists argue that such effects could still be explained within their framework, no complete joint fit including BAO, CMB, and SNe has been published. Because ΛCDM naturally reproduces these correlations through a single set of parameters, this coherence strongly favors the standard model.

The cosmos's surveyed near-homogeneity is precisely why ΛCDM performs so smoothly. It assumes an FLRW metric with small perturbations, consistent with the CMB, BAO isotropy, and the uniformity of the Hubble flow. A single expansion history $H(z)$ fits supernovae, BAO, and CMB data without invoking any regional lapse in clock rates or large-scale curvature gradients. Timescape, by contrast, requires large deviations from local homogeneity—deviations that would make distinct marks on multiple independent types of observed phenomena. Their absence therefore strongly favors the ΛCDM framework, whose internal coherence and observational consistency remain unmatched among competing cosmological models.

9. Conclusion

The Timescape cosmology provides a compelling thought experiment: it demonstrates how differential expansion and gravitational time dilation could, in principle, produce an apparent acceleration without dark energy. However, its success is primarily internal—rooted in the mathematical construction of its two-region framework rather than in measured properties of the cosmic web. While supernova data alone cannot exclude Timescape, the model struggles in finding a self-consistent multi-probe fit for CMB, BAO, and SNe data.

Given the precision and consistency with which ΛCDM reproduces the universe's expansion history, it remains the most reliable framework for explaining cosmic acceleration.

References

DESI Collaboration (2025). Desi dr2 results ii: Measurements of baryon acoustic oscillations and cosmological constraints. *arXiv e-prints*. DESI Collaboration, 2025, arXiv:2503.14738.

Ishibashi, A. and Wald, R. M. (2006). Can the acceleration of our universe be explained by the effects of inhomogeneities? *Classical and Quantum Gravity*, 23(1):235–250.

Nazer, M. A. and Wiltshire, D. L. (2015). Cosmic microwave background anisotropies in the timescape cosmology. *Physical Review D*, 91:063519.

Planck Collaboration (2020). Planck 2018 results. vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6.

Seifert, E., Wiltshire, D., Petrosian, M., et al. (2024). Supernovae evidence for foundational change to cosmological models. *arXiv e-prints*.

Smale, P. R. and Wiltshire, D. L. (2011). Supernova tests of the timescape cosmology. *Monthly Notices of the Royal Astronomical Society*, 413(1):367–385.

Weinberg, D. H., Mortonson, M. J., Eisenstein, D. J., Hirata, C., Riess, A. G., and Rozo, E. (2013). Observational probes of cosmic acceleration. *Physics Reports*, 530:87–255.

Wiltshire, D. L. (2007). Exact solution to the averaging problem in cosmology. *New Journal of Physics*, 9:377.

Wiltshire, D. L. (2008). Gravitational energy as dark energy: Observational tests of the timescape cosmology. *Astrophysical Journal Letters*, 672(2):L91–L94.